

Rafael Mendonça Duarte

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Profile Summary

Information Systems student at the Institute of Mathematical and Computer Sciences of the University of São Paulo (ICMC-USP) with a strong academic record (GPA 8.6/10, no failures). Currently dedicated to the development and improvement of Graph Convolutional Neural Networks (GCNs) applied to image classification, achieving State-of-the-Art (SOTA) performance in recent research. Experienced in High-Performance Computing, deep learning, software integration, and business process automation.

Education

Bachelor of Science in Information Systems, University of São Paulo (USP) – Jan 2025 – Dec 2028 (Expected)
ICMC

Academic Standing: GPA: 8.6/10 (No failed courses).

Relevant Coursework: Data Structures, Modern Operating Systems, Object-Oriented Programming (OOP), Calculus, Linear Algebra, Statistics.

High School, Anglo Alante Jundiaí 2022 – 2024

Research Experience

Scientific Initiation Student / Research Intern, University of São Paulo (USP) Aug 2026 – Present

Funded by FAPESP – São Paulo Research Foundation (Grant #2026/16921-8). Advisor: Prof. Dr. Lucas Pascotti Valem.

- Conducting research on the project "*Identification of Representative Vertices and Feature Graph Fusion in Graph Convolutional Networks*".
- Investigating graph fusion strategies to effectively combine complementary information from different feature extractors while minimizing noise in Graph Convolutional Networks (GCNs) applied to image classification.
- Exploring the insertion and identification of representative (class prototypes) and synthetic (image-cluster-based) vertices to improve graph topology and support model generalization under scarce data or limited labels.

Key Skills: Graph Neural Networks (GNNs), PyTorch, Machine Learning, Data Structures and Data Science.

Scientific Initiation Student / Research Intern, University of São Paulo (USP) Nov 2025 – Jul 2026

Funded by "Pró-Reitoria de Pesquisa e Inovação". Advisor: Prof. Dr. Lucas Pascotti Valem.

- Investigated the influence of structural graph properties and message aggregation strategies in GCNs applied to image classification with limited labeled data.
- Developed **GRaND**e (Gaussian Rank-based Neighborhood Degree), a novel adaptive centrality measure that achieved **State-of-the-Art (SOTA)** performance, **improving accuracy by up to 14%** over existing baselines.
- **First-authored** the paper introducing the proposed method, which was accepted for publication at the Brazilian Symposium on Databases (**SBBD 2026**).

Key Skills: Graph Neural Networks (GNNs), PyTorch, Machine Learning, Data Structures and Data Science.

Professional Experience

Technical IT Monitor, Pró-Aluno Program (USP) Sep 2025 – Oct 2025

Funded by "Pró-Reitoria de Graduação", "Pró-aluno Program".

- Provided software and hardware technical support to users in the ICMC undergraduate laboratories.
- Assisted students in the use of equipment and ensured the proper functioning and care of the local IT infrastructure.

IT Management Intern, FK Trading Jul 2025 – Sep 2025

- **Automation & Tooling:** Developed custom **Node.js** scripts for API integrations and web scraping to automate

CURRICULUM VITAE – **Rafael Mendonça Duarte**
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lead generation and client follow-ups. Implemented WhatsApp chatbots and trained 3 out of 5 internal coworkers, replacing manual Kanban funnels and boosting the team's **operational efficiency by 30%**.

- **Technical Consulting:** Acted as a consultant for 15 corporate clients, mapping business workflows and standardizing processes to guide their first-time adoption of CRM and ERP systems (RD Station, OMIE).
- **Key Achievement:** Optimized a B2B client's sales pipeline to seamlessly manage up to **1,000 leads/day** and drive 200 monthly sales. In one critical instance, leveraged automations to help a 5-person sales team clear a backlog of **800 stalled leads in just 4 days**.

Key Skills: Node.js, API Integration, CRM/ERP (RD Station, OMIE), Business Process Automation, Web Scraping, Mentorship.

Extension & Leadership Projects

Director of High-Performance Computing (HPC) / Coordinator, Project Egrégora (USP) Jun 2025 – Present

IEEE Computer Society Chapter & EESC - São Carlos School of Engineering.

- Acting as Student Project Coordinator for the technological extension area aimed at the re-characterization of "TV Box" hardware devices.
- Coordinated the assembly of low-cost distributed computing clusters and the implementation of local database servers.
- We orchestrated a **performance and scalability evaluation** of databases on resource-constrained ARM hardware utilizing Docker environments and, we analyzing the latency and throughput trade-offs across different architectures, benchmarking direct DB access with a focus to **optimize I/O problems**.

Key Skills: Docker, Database Architecture, Performance Benchmarking, System Administration and FastAPI.

Extension Member, Corvus AI Group (USP) Mar 2026 – Present

Project: Combating Digital Disinformation: Fake Image Detection.

- Acting as a volunteer member researching deep learning techniques in computer vision.
- Contributed to the development and continuous optimization of algorithms aimed at detecting fake images and mitigating digital disinformation.

Teaching Coordinator Member, DATA – Data Science Study Group (USP) Mar 2026 – Present

ICMC – Institute of Mathematical and Computer Sciences, University of São Paulo.

- Member of the Teaching Team Coordination of the DATA Group, responsible for planning and delivering introductory Data Science lectures.
- Taught **4 lectures** on **Introduction to Data Science** and **scikit-learn** to a class of **10 students**, applying hands-on analysis over a public Kaggle dataset of **FIFA World Cup 2026** match data.

Key Skills: scikit-learn, Data Science, Machine Learning, Teaching.

Digital Literacy Instructor for Seniors, "Informática para a Terceira Idade" (USP) Feb 2026 – Jun 2026

- 3rd Edition of the extension activity aimed at the digital inclusion of the external community of São Carlos.
- Facilitated classes for the elderly community as a student monitor, teaching essential computer skills, internet navigation, and individually clarifying doubts to promote **technical autonomy**.

Publications

- **DUARTE, R. M.; PONCIANO, J. R.; VALEM, L. P.** "Gaussian Rank-Based Neighborhood Degree for Graph Neural Networks in Image Classification." In: **Proceedings of the 41st Brazilian Symposium on Databases (SBBD 2026)**, São Carlos, SP, Brazil, 2026, pp. 575–588.
 - **Status:** Published in Conference Proceedings (Qualis A4).
 - **DOI (Preprint):** <https://doi.org/10.48550/arXiv.2605.24367>
 - **Relevance:** The main contribution is the GRaNDe method, a dynamic centrality measure that integrates neighborhood analysis with Gaussian distance weighting to capture node importance, effectively improving GNNs in image classification.
- **BRITO, G. M.; DUARTE, R. M.; VALEM, L. P.** "Rank Correlation Graphs with Mutual Neighborhood Weighting for

Graph Convolutional Networks in Image Classification." In: **Pattern Recognition Letters**, 2026.

- **Status:** Published (Qualis A1).
 - **DOI:** <https://doi.org/10.1016/j.patrec.2026.09.012>
 - **Relevance:** Introduces a strategy based on mutual neighborhood for weighting correlation graphs, bringing a comparative analysis of different weighting functions and topologies for consistent image classification.
- **DUARTE, R. M.; BRITO, G. M.; VALEM, L. P.** "*Rank Correlation Meets Gaussian Degree Normalization in GNNs for Image Classification.*" In: **Proceedings of the 39th SIBGRAPI Conference on Graphics, Patterns and Images (SIBGRAPI 2026)**, Goiânia, GO, Brazil, 2026.
 - **Status:** Accepted for publication, Main Track (Qualis A3).
 - **Relevance:** Proposes D-GRaNDe, unifying the Density-Guided Correlation Graph (DGCG) construction with GRaNDe degree normalization, showing that rank-correlation graphs and dynamic degree normalization are complementary and jointly outperform either component alone.
 - **BENATO, B. C.; VALEM, L. P.; BRITO, G. M.; BEZERRA, I. H. C.; DUARTE, R. M.; NONATO, L. G.** "*An Introduction to Graph Neural Networks in Computer Vision: A Practical Perspective.*" In: **Proceedings of the 39th SIBGRAPI Conference on Graphics, Patterns and Images (SIBGRAPI 2026), Tutorial Track**, Goiânia, GO, Brazil, 2026.
 - **Status:** Accepted for publication, Tutorial Track (Qualis A3).
 - **Relevance:** Practical tutorial covering GNN foundations and their application to image classification, retrieval, and segmentation, including the GCNVis visualization tool for inspecting visual graphs.

Awards & Distinctions

- **Academic Excellence Award (2024)** – Granted by Anglo Alante Jundiaí.
- **Bronze Medal (2024)** – Kangaroo Math Competition.
- **Bronze Medal (2024)** – Brazilian Astronomy and Astronautics Olympiad (OBA).
- **Silver Medal (2023)** – Kangaroo Math Competition.
- **Best Position Paper & Best Director (2023)** – High School United Nations Human Rights Council 1 (FACAMP) / FAMUN & CJONU. Highlighted strong communication, diplomacy, and research skills.
- **Bronze Medal (2022)** – Brazilian Rocket Exhibition (MOBFOG).
- **Two-time Finalist (4th Phase)** – National History Olympiad (ONHB).

Technical Skills

Programming & Web: Python, Java, Node.js, SQL, FastAPI.

Machine Learning & Data Science: PyTorch, Graph Neural Networks (GNNs), NLP (Word2Vec), XGBoost, Pandas.

Tools & Infrastructure: Docker, Git, Linux, API Integration.

Core Concepts: Data Structures, Object-Oriented Programming (OOP), Database Architecture, Business Process Automation, Statistics.

Languages

Portuguese: Native

| **English:** Intermediate – proficient in technical reading and writing